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COMPUTER

CHAPTER 29

Mobile Phones

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CHAPTER 29

Mobile Phones

Computer Awareness

IN THIS CHAPTER

- Cellular networks
- Mobile-phone milestones
- Identification codes
- Call routing
- Wireless technologies
- Mobile generations

- DuckDuckGo = Privacy-focused.
- Safari = Browser, not Search Engine.
- Archie (1990) = First Internet Search Engine.

A Mobile Phone (Cell Phone) is an electronic communication device used to make and receive calls using radio frequency signals (electromagnetic waves) while the user is moving within a cellular network coverage area.

Mobile phones communicate through a network of specialized base stations called Cell

Sites (Mobile Towers).

A mobile phone is an electronic device used for mobile telecommunications over a cellular network, allowing users to communicate through voice calls, messages, and internet services while moving from one location to another.

Key Features of Mobile Phones

Modern mobile phones support:

- Voice Calls
- SMS (Short Message Service)
- MMS (Multimedia Messaging Service)
- Email
- Internet Access
- Bluetooth
- Camera & Video Recording
- GPS Navigation
- Music Player
- Radio
- Mobile Banking
- Social Media Applications

Exam Point

Because mobile phones use a cellular network architecture, they are also called:

- Cellular Phones
- Cell Phones

IMPORTANT INVENTORS AND HISTORICAL FACTS

Invention	Inventor/Organization	Year
Walkie-Talkie	Donald Hings	1937
First handheld cellular-phone prototype	Martin Cooper	1973
First commercially available handheld cellular phone	Motorola DynaTAC 8000X	1983
First Smartphone	IBM Simon Personal Communicator	1994
First Cellular Network	Nippon Telegraph and Telephone (NTT), Japan	1979

Exam Point :Martin Cooper is known as the "Father of the Mobile Phone".

HOW MOBILE PHONES WORK



Simplified cellular call path

Mobile phones communicate using Radio Waves.

A mobile phone converts voice into electrical signals and transmits them to nearby cell towers using radio frequencies.

CELLULAR NETWORK CONCEPT

A large geographical area is divided into small regions called Cells.

Each cell is served by a base station (cell tower).

Shape of Cell

For network planning purposes, the cell is assumed to be HEXAGON

Full Duplex Communication

Mobile phones support Full Duplex Communication.

Both users can speak and listen simultaneously.

MULTIPLE ACCESS TECHNIQUES

To allow many users to communicate simultaneously, cellular systems use multiple access techniques.

1. FDMA (Frequency Division Multiple Access)

Each user gets a separate frequency channel.

Features

- Different frequencies assigned to different users.
- Neighboring cells use different frequencies.

Example

User A → Frequency 1

User B → Frequency 2

User C → Frequency 3

2. TDMA (Time Division Multiple Access)

Multiple users share the same frequency but at different time slots.

Features

- Frequency divided by time.
- Each user transmits in a specific time period.

3. CDMA (Code Division Multiple Access)

All users share the same frequency but use unique codes.

Features

- More efficient than FDMA.
- Better capacity and security.

4. PDMA (Polarization Division Multiple Access)

Uses different polarization states of radio waves to separate signals.

MOBILE PHONE IDENTIFICATION CODES

The following ESN, MIN and SID codes describe legacy cellular systems; IMEI is used to identify modern cellular equipment.

1. ESN (Electronic Serial Number)

- Unique 32-bit number
- Permanently programmed into the phone

Exam Point

ESN is permanent.

2. MIN (Mobile Identification Number)

- 10-digit number
- Derived from the mobile number

3. SID (System Identification Code)

- 15-bit identifier, often shown using up to five decimal digits
- Assigned to each cellular carrier

Example

Legacy cellular-network identifiers; these are not the standard subscriber identifiers used by modern Jio, Airtel or Vi networks.

4 The IMEI (International Mobile Equipment Identity)

IMEI is a 15-digit identifier assigned to cellular equipment; it does not apply to every type of mobile device.

This number is crucial for several reasons:

- Device Identification .
- Network Security: Mobile networks use IMEI to allow or block devices from accessing

the network.

- Theft Prevention: If a phone is lost or stolen, the IMEI can be blacklisted, preventing

its use with any SIM card.

- Warranty and Service
- You can usually find your IMEI by dialling *#06# on the keypad or checking the

device’s settings and packaging.

COMPARISON OF MOBILE CODES

Code	Full Form	Purpose
ESN	Electronic Serial Number	Identifies Phone
MIN	Mobile Identification Number	Identifies Subscriber
SID	System Identification Code	Identifies Network Provider

SIM CARD

SIM stands for: Subscriber Identity Module

A SIM card is a smart card issued by a mobile service provider(AIRTEL,JIO) that stores subscriber information.

Information Stored in SIM

- IMSI
- Phone Number
- Authentication Information
- Network Details

IMSI

Full Form :International Mobile Subscriber Identity

A unique number stored in the SIM card used to identify the subscriber.

TMSI

Full Form :Temporary Mobile Subscriber Identity

After IMSI verification, the network assigns a temporary identity called TMSI.

Purpose

- Improves privacy
- Protects IMSI from exposure

ESIM

Full Form :Embedded Subscriber Identity Module

Features

- Built into device hardware
- No physical SIM card required
- Can be activated digitally

Exam Point

eSIM = Embedded SIM.

WHAT HAPPENS WHEN WE MAKE A CALL?

Step 1 Phone searches for a nearby control channel.

If unavailable:

"No Service" is displayed.

Step 2

Phone checks SID.

If SID matches:

Phone identifies its home network.

Step 3

Phone sends a registration request to the network.

Step 4

MTSO(Mobile telephone switching office) records the phone's location.

Step 5

MTSO assigns a frequency pair.

Step 6

Call is connected.

Step 7

If user moves into another cell:

Network performs Handoff (Handover). Call continues without interruption.

MTSO (Mobile Telephone Switching Office)

Functions

- Call Routing
- Location Tracking
- Frequency Allocation
- Handover Management

Exam Point :MTSO is the central controller of cellular communication.

HANDOFF (HANDOVER)

The process of transferring an ongoing call from one cell tower to another without disconnecting the call.

Example: While traveling in a car, your call continues because handoff occurs automatically

IMPORTANT MOBILE COMMUNICATION TERMS

Term	Full Form
Wi-Fi	Wireless LAN brand; “Wireless Fidelity” is a common exam-guide expansion, not a technical acronym.
WiMAX	Worldwide Interoperability for Microwave Access
WAP	Wireless Application Protocol
GPS	Global Positioning System
WCDMA	Wideband Code Division Multiple Access
GSM	Global System for Mobile Communications
LTE	Long Term Evolution
VoLTE	Voice over LTE
CDMA	Code Division Multiple Access

GSM SYSTEM

Full Form :Global System for Mobile Communications

Features

- Most widely used mobile communication standard
- Digital system
- Considered a 2G technology

GSM CHANNELS

Control Channel :Used for network management and signaling.

Traffic Channel :Used for voice and data transmission.

OPERATING SYSTEMS IN MOBILE PHONES

ANDROID

Based On :Linux Kernel

Developed By :Google

IOS

Developed By :Apple Inc.

Based On :Darwin Operating System

Exam Point

Android → Linux Kernel

iOS → Darwin

INTERNET SERVICE PROVIDERS (ISP) Telenet

Early commercial packet-switched network; describing Telenet as the first modern Internet service provider is inaccurate.

Established in 1974.

VSNL (Videsh Sanchar Nigam Limited)

VSNL launched public commercial Internet services in India on 15 August 1995.

Exam Point :VSNL = First Internet Service Provider in India.

PHUBBING

Ignoring a person physically present with you because you are busy using your smartphone.

GENERATIONS OF MOBILE NETWORKS

Evolution of Mobile Communication

0G 2.75G

| |

1G 3G

| |

2G 4G

| |

2.5G 5G

0G (ZERO GENERATION)

Features

- Pre-cellular mobile communication
- Radio Telephones
- Installed in vehicles

Services :Voice only

1G (FIRST GENERATION)

Introduced :1979

Country :Japan

Organization :NTT(Nippon telephone and telegraph)

Features

- Analog communication
- Voice only
- Low quality

1G used analog voice; 2.4 kbps is not a universal digital data rate for analog 1G networks.

Exam Point: 1G = Analog Technology

2G (SECOND GENERATION)

Features

- Digital technology
- Voice Calls
- SMS
- Email

Example rate: 64 kbps; 2G data rates vary by technology and configuration.

Exam Point :2G introduced SMS.

2.5G

Technology :GPRS (General Packet Radio Service)

Features

- Packet-switched data
- Internet Access

2.75G

Technology :EDGE (Enhanced Data Rates for GSM Evolution)

Example operating rate: 128 kbps. EDGE can support higher rates; 128 kbps is not its universal maximum.

Features

- Faster Internet

3G (THIRD GENERATION)

Features

- High-Speed Internet
- Video Calling
- Multimedia Services

Technologies

- UMTS
- WCDMA

Early 3G targets included up to 2 Mbps under favourable low-mobility conditions; later 3G systems achieved higher rates.

Exam Point :3G introduced practical mobile internet and video calling.

4G (FOURTH GENERATION)

Features

- IP-based Network

- High-speed broadband
- HD Streaming
- Online Gaming

Technologies

- LTE (Long Term Evolution)
- VoLTE(Voice over LTE)

100 Mbps is an early high-mobility 4G target; actual and peak rates vary by system.

Exam Point :VoLTE supports voice and data simultaneously.

5G (FIFTH GENERATION)

Features

- Ultra-high speed
- Low latency
- Massive device connectivity
- Enhanced security

Applications

- Smart Cities
- IoT (Internet of Things)
- Self-driving Cars
- Remote Surgery
- Artificial Intelligence

Speed :Can exceed 1 Gbps under ideal conditions.

Exam Point

5G can be much faster than 4G, but there is no fixed 10–30-fold improvement across all networks.

MOST IMPORTANT EXAM FACTS

1.Martin Cooper demonstrated a handheld cellular-phone prototype in 1973.

2.Motorola DynaTAC 8000X was the first commercially available handheld cellular phone.

3.IBM Simon was the first smartphone.

4.Hexagon is the ideal cell shape.

5.Mobile phones use radio waves.

6.Google Android is based on Linux Kernel.

7.iOS is based on Darwin.

8.GSM stands for Global System for Mobile Communications.

9.VSNL was India's first ISP.

10.Telenet was an early commercial packet-switched network established in 1974.

11.IMSI is stored in the SIM card.

12.TMSI is a temporary identity.

13.MTSO controls call switching.

14.Handoff transfers calls between cells.

15.1G = Analog.

16.2G = Digital + SMS.

17.3G = Mobile Internet. 18.4G = LTE & VoLTE.

19.5G = Ultra-fast communication.

20.Phubbing = Phone + Snubbing.

QUICK REVISION NOTES

- Mobile phones use radio waves.
- Hexagon is the shape of a cellular cell.
- Martin Cooper invented the mobile phone.
- Motorola DynaTAC 8000X = First commercial mobile phone.
- IBM Simon = First smartphone.
- Android = Linux Kernel.
- iOS = Darwin.
- SIM = Subscriber Identity Module.
- IMSI = International Mobile Subscriber Identity.
- eSIM = Embedded SIM.
- GSM = Global System for Mobile Communications.
- Wi-Fi = Wireless LAN brand name; Wireless Fidelity is a commonly used exam-guide expansion, not a technical full form..
- GPS = Global Positioning System.
- MTSO controls mobile call routing.
- Handoff transfers calls between cells.
- 1G = Analog.
- 2G = SMS.
- 3G = Mobile Internet.
- 4G = LTE.
- 5G = Ultra-fast connectivity.